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SpikeTrack: High-performance and Energy-efficient Event-Based Object Tracking with Spiking Neural Network

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Abstract

*Event cameras have attracted considerable attention for object tracking due to their microsecond-level temporal resolution and wide dynamic range, yet effectively harnessing spiking neural networks (SNNs) in this domain remains challenging. In this paper, we introduce **SpikeTrack**, a purely spike-driven framework for single-object tracking that addresses the shortcomings of RGB-based approaches in fast-motion or target appearance change. Central to SpikeTrack is the **M**ulti-**S**earch-sequence-and-**S**ingle-**T**emplate (MSST) training paradigm, which captures rich temporal dependencies, alongside a **D**ynamic **I**nteger **L**eaky **I**ntegrate-and-**F**ire (DI-LIF) neuron that adaptively predicts integer-valued activations based on the input features during training and converts them into spikes during inference. Our design preserves the intrinsic sparsity and fine-grained spatiotemporal acuity of event data, resulting in efficient energy consumption without sacrificing performance. Extensive evaluations on FE108, FELT, and VisEvent demonstrate that SpikeTrack exceeds the performance of state-of-the-art trackers in both accuracy and efficiency. Furthermore, ablation studies validate each module's contribution, highlighting the practical potential of spike-driven architectures for future vision applications.*

1. Introduction

Object tracking is a foundational task in computer vision, underpinning a variety of applications ranging from surveillance to autonomous navigation and robotics. While recent RGB-based methods have achieved noteworthy progress across numerous tracking benchmarks [4, 5, 19, 25, 28, 35, 36], they often exhibit limitations in high-speed or high-dynamic-range environments, thereby impeding their reliability in real-world deployments. By contrast, event cameras capture scene changes at microsecond-level temporal resolution and offer a substantially broader dynamic range (e.g., 140 dB vs. 60 dB), enabling them to perceive rapid motion cues and extreme lighting variations that traditional sensors might overlook. Consequently, event-based sensors offer significant opportunities for advancing tracking, especially under challenging conditions.

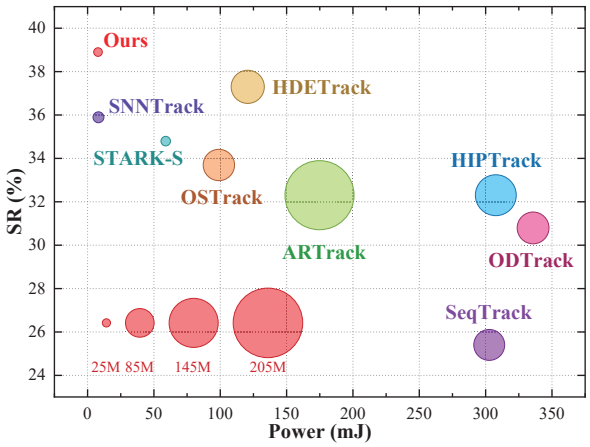


Figure 1. Comparison of SpikeTrack with other trackers on VisEvent in SR, energy consumption and model size.

While certain event-based trackers [22, 32] have effectively utilized insights from RGB tracking approaches, like correlation filters and Siamese networks, these approaches often overlook the significant temporal cues inherent in events. The temporal information is essential in tracking as it conveys the state transitions of targets between frames. Recent studies [33, 34] have therefore incorporated temporal processing mechanisms, including Long Short-Term Memory networks (LSTMs) and Spiking Neural Networks (SNNs). Among them, the potential accumulation, decay, and resting process in SNNs render it a natural temporal feature extractor. Moreover, the discrete pulse-based communication between neurons in SNNs offers low energy consumption and aligns well with the sparsity and asynchrony of event data, thus presenting great potential for event-based

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tracking. For example, STNet [33] introduces a three-layer SNN module featuring a dynamic spiking threshold to improve the extraction of temporal cues. SNNTrack [34] focuses on the impact of the adaptive membrane time constant in SNN on tracking performance. Despite these methods demonstrate effectiveness, they typically employ a CNN-SNN hybrid architecture, which relies heavily on the feature extraction capabilities of CNN and does not fully leverage the low energy consumption benefits of SNN. While pure SNN architectures have been extensively studied in classification tasks, their effectiveness in event-based tracking remains under-researched.

Recently, Spiking Transformer architectures have shown promising progress but still struggle to match the performance of ANNs [37]. To address this, Luo et al. [14] proposed activating integer values during training and maintaining spike-driven processing at inference by extending the virtual time steps. However, this approach adopts a fixed maximum spike count across all scenarios, lacking adaptability and thus limiting its effectiveness in handling the dynamic motion and appearance variations inherent in object tracking tasks.

This paper presents a novel pure spike-SNN architecture designed for event-based object tracking tasks. Our method adopts the popular Spiking Transformer [14, 27] as the backbone network. To enhance adaptation to the tracking task, we propose two essential components: (i) the first framework that leverages SNNs' inherent temporal dynamics to model inter-frame sequential evolution of target trajectories through a **M**ulti-**S**earch-sequence-and-**S**ingle-**T**emplate (MSST) training paradigm, eliminating the need for auxiliary temporal processing modules in object tracking tasks; (ii) We propose a **D**ynamic **I**nteger **L**eaky **I**ntegrate-and-**F**ire (DI-LIF) neuron that adaptively predicts integer activations based on input features during training and converts them into spikes during inference. Under dense event input, it increases the maximum emitted integer value to enhance responsiveness to informative cues, while under sparse input, it reduces the value to suppress redundancy. This strategy effectively mitigates quantization error of the conversion between membrane potential and spikes while preserving the discrete nature of SNNs.

Extensive experiments on the FE108 [31], FELT [21], and VisEvent [20] datasets validate the effectiveness of the proposed SpikeTrack. Figure 1 illustrates that our approach attains the optimal trade-off among parameter count, energy consumption, and accuracy. Ablation experiments evidence the importance of each key component of SpikeTrack. It is worth noting that our method is the first work to present a pure SNN architecture to event-based object tracking, which fully exploits the low energy consumption and spatiotemporal modeling capabilities of SNN.

In summary, we make the following contributions:

- We propose a novel pure spike-SNN architecture with MSST training paradigm for event-based object tracking, allowing us to effectively capture the historical temporal cues of the target motion and appearance changes.
- We propose a DI-LIF neuron integrated into the Transformer module, which adaptively adjusts the maximum emitted integer value of SNNs based on input features, enabling sample-adaptive quantization and efficient temporal feature encoding.
- Extensively experimental results validate that our approach outperforms state-of-the-art methods. Our ablation study evidences the effectiveness of each key component of the proposed SpikeTrack.

2. Background and Related Work

2.1. I-LIF Spiking Neuron

One recent development in SNNs is the introduction of the I-LIF neuron model, which augments the traditional LIF mechanism and thereby broadens the representational capacity of SNNs [14, 24, 27]. By incorporating an intrinsic parameterization, I-LIF can deliver performance rivaling that of standard artificial neural networks (ANNs) while retaining a fundamentally spike-driven paradigm.

In training, the I-LIF neuron typically encodes information through a firing-rate representation:

$$s^{\ell} = \frac{1}{D} \cdot \left\lfloor \mathrm{clip}\big(x^{\ell}, 0, D\big) \right\rfloor, \tag{1}$$

where x^{ℓ} is the input current at the ℓ-th layer, D is the maximum emitted integer value, and $\mathrm{clip}(\cdot)$ restricts values to the interval $[0, D]$. $\lfloor \cdot \rfloor$ denotes the floor operation. During inference, the neuron's firing rate s^{ℓ} is decomposed into D binary events:

$$s^{\ell} = \frac{1}{D} \sum_{t=1}^{D} s^{\ell}[t], \tag{2}$$

where each $s^{\ell}[t]$ is a discrete spike (0 or 1). Subsequent layers receive these spikes either through linear or elementwise (Hadamard) operations, formulated as

$$x^{\ell+1} = \sum_{t=1}^{D} w^{\ell} \otimes s^{\ell}[t], \tag{3}$$

where w^{ℓ} is the weight parameter at layer ℓ, and $\otimes$ denotes multiplication (e.g., matrix multiply or convolution). This architectural choice maintains the core spike-based principle of SNNs while enabling multiply-accumulate (MAC) operations to be replaced with more efficient accumulative (AC) operations, leading to reductions in model size and computational overhead—an attractive property for edge-level or resource-constrained deployments.

2.2. Event-based Object Tracking

Multiple studies [3, 22, 32] suggest that traditional RGB-based single-object tracking (SOT) methods often perform poorly when confronted with event data, largely due to the scarcity of texture cues essential for conventional trackers. In contrast, SNNs exhibit a spike-driven operating mechanism aligned with the asynchronous nature of event camera outputs, pointing to an intrinsic compatibility between SNNs and event streams. This connection has motivated several works [10, 33, 34] to leverage SNNs for event-based tracking or for deploying tracking algorithms on resource-constrained edge devices. Early SNN-based efforts, such as STNet [33] and SNNTrack [34], demonstrate promising results by extracting temporal features, yet they generally integrate hybrid SNN-ANN designs and do not exploit self-attention for more robust template-search interaction. Consequently, the potential for energy-efficient yet high-performing solutions inherent in SNNs remains underutilized. To address these limitations, we introduce a pure spike-SNN architecture designed for event-based object tracking, effectively leveraging the low-power and spike-driven processing benefits of SNNs.

2.3. Historical Temporal Tracking

Leveraging historical cues significantly enhances single-object tracking by capturing temporal target evolution and improving robustness to fast motion, occlusion, and deformation. Existing methods typically fall into two categories. The first utilizes single search frames with dynamic kernels or update branches to adapt to appearance changes [1, 7, 9, 18], but often depends on hand-crafted rules, limiting generalization. The second employs multiple search frames in Transformer-based architectures to explicitly model temporal dynamics [17, 36], yet incurs high computational cost and limited long-range performance. In contrast, we leverage the inherent timing characteristics of SNNs and propose a MSST training scheme. By distributing search frames across time while sharing a single reference template, our method naturally captures inter-frame dynamics and handles significant appearance changes. Furthermore, the sparse and memory-efficient nature of SNNs enables robust tracking over extended sequences without manual update mechanisms or large recurrent modules.

3. SpikeTrack

The overall architecture of our proposed SpikeTrack is illustrated in Figure 2, which is an end-to-end tracker and does not require any data augmentation or post-processing. We first divide the given event stream into a n-bin voxel grid over a period of time. For each bin, an event frame can be aggregated by accumulating events over time, and each frame contains two channels of positive and negative polarities. To strengthen the temporal coherence modeling of the tracker, we design a sequential input pipeline where multiple search frames (captured at consecutive timesteps) and the template frame are jointly processed by the I-LIF based SNN Conv Module as temporally ordered spike trains. In addition, the membrane potential $\mathbf{U}$ of the spiking neurons is updated between time intervals. Next, the features are input into our DI-LIF based SNN Transformer Module. Our DI-LIF can dynamically and adaptively adjust the maximum emitted integer value D according to the input features. Finally, the cross-correlation features between template and search features are fed into the I-LIF based SNN Tracking prediction head to locate the targets.

3.1. Events Representation and MSST Training

Event cameras asynchronously trigger pixel-wise events when logarithmic brightness changes exceed a contrast threshold C, governed by the inequality $L(x, y, t) - L(x, y, t - \Delta t) \geq p \cdot C$, where $p \in \{-1, +1\}$ encodes the polarity of brightness variation. Following [30, 33], to process asynchronous event streams $\mathcal{E} = \{(x_k, y_k, t_k, p_k)\}_{k=1}^{N}$ with SNNs, we aggregate events into spatiotemporal representations through a polarity-aware encoding strategy. Positive and negative events are accumulated into separate channels via spatial-temporal Dirac delta mapping:

$$E^{\pm}(x, y, t) = \sum_{k} \delta(x - x_k, y - y_k) \cdot \delta(t - t_k) \cdot \mathbb{I}(p_k = \pm 1), \tag{4}$$

where $\mathbb{I}(\cdot)$ filters events by polarity. These polarity-specific channels are concatenated into a dual-channel tensor $I_t \in \mathbb{R}^{H \times W \times 2}$, forming the unified input of our tracking network. All baseline methods utilize identical event encoding to ensure fair benchmarking.

In typical tracking architectures, the input consists of a template frame and a single search frame, which limits the ability to model temporal information. To address this, we adopt a MSST training paradigm. This scheme aligns well with the strengths of SNNs, which naturally encode spatiotemporal patterns through membrane potential accumulation over time. Additionally, object tracking inherently relies on temporal continuity and motion consistency. Therefore, using multiple search frames provides richer temporal context, enabling more robust and accurate localization across dynamic scenes.

3.2. I-LIF Based SNN Conv Module

Convolutional SNN architectures based on I-LIF neurons have been effectively applied to enhance object detection performance by extending the virtual time steps to suppress quantization errors and convert them into effective spikes during inference. Inspired by [27], as illustrated in Figure 2(b), we cascade four I-LIF-based SNN convolutional

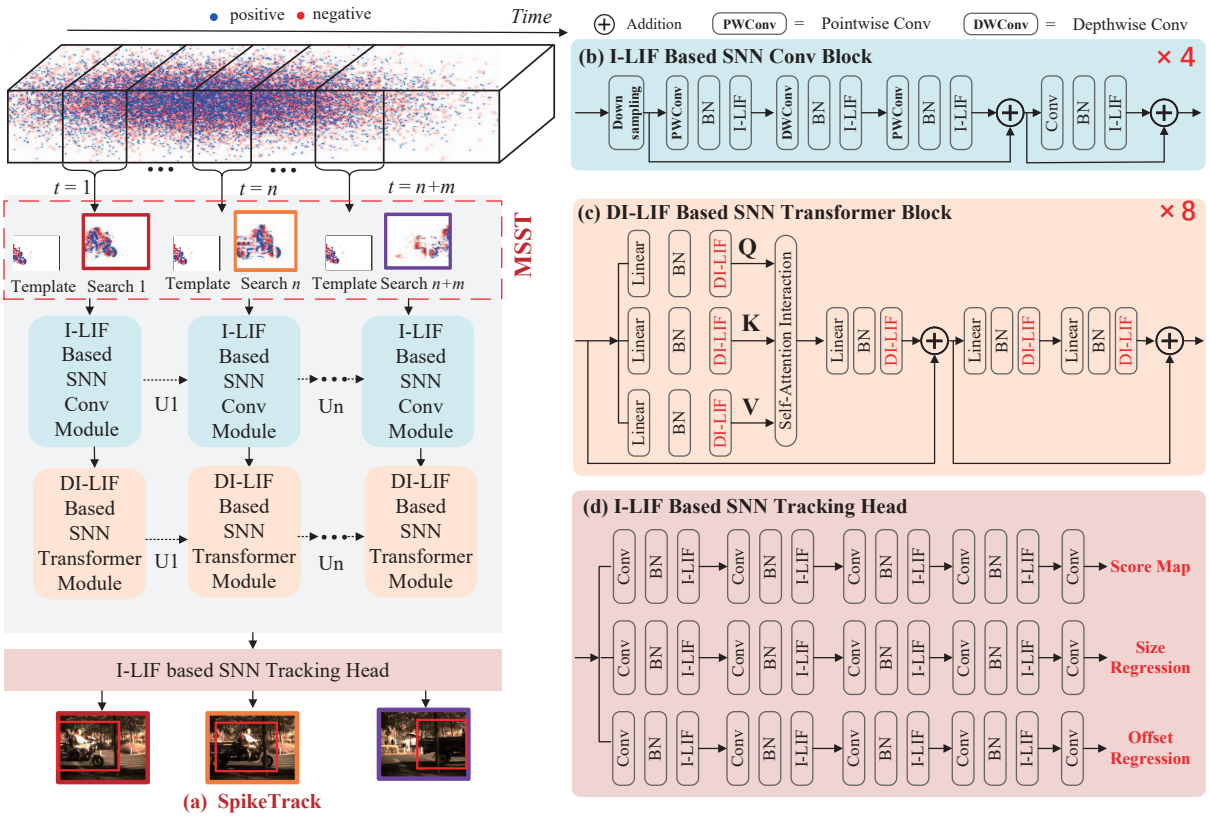


Figure 2. (a) Details of our SpikeTrack, which employs the Multi-Search-sequence-and Single-Template (MSST) training paradigm to capture the temporal cues related to target state changes; and three components of our SpikeTrack: (b) I-LIF Based SNN Conv Block, (c) DI-LIF Based SNN Transformer Block and (d) I-LIF Based SNN Tracking Head.

blocks to extract spatiotemporal features from both the template and search event frames.

Each proposed block implements hierarchical feature refinement through spiking operator fusion, formulated as:

$$\begin{aligned} \mathbf{U}' &= \mathbf{U} + \mathrm{SNNSepConv}(\mathbf{U}), \\ \mathbf{U}'' &= \mathbf{U}' + \mathrm{SNNConvGroup}(\mathbf{U}'), \\ \mathrm{SNNConvGroup}(\mathbf{U}') &= \mathrm{Conv}(\mathcal{SN}(\mathrm{Conv}(\mathcal{SN}(\mathbf{U}')))), \end{aligned} \tag{5}$$

where $\mathcal{SN}(\cdot)$ denotes the I-LIF spiking neuron layer; $\mathrm{SNNSepConv}(\cdot)$ decomposes as:

$$\begin{aligned} \mathrm{SNNSepConv}(\mathbf{U}) &= \mathrm{Conv}_{\mathrm{pw}}(\mathcal{SN}(\mathbf{V}_2)), \\ \mathbf{V}_2 &= \mathrm{Conv}_{\mathrm{dw}}(\mathcal{SN}(\mathbf{V}_1)), \\ \mathbf{V}_1 &= \mathrm{Conv}_{\mathrm{pw}}(\mathcal{SN}(\mathbf{U})), \end{aligned} \tag{6}$$

where $\mathrm{Conv}_{\mathrm{pw}}$ and $\mathrm{Conv}_{\mathrm{dw}}$ represent pointwise and depthwise convolutions [6], respectively; batch normalization is omitted for notation simplicity.

3.3. DI-LIF Based SNN Transformer Module

In the I-LIF neuron model, a predefined hyperparameter D is used to constrain the maximum emitted integer value. However, such a fixed setting may hinder tracking performance due to the dynamic nature of the task, where target motion speed and appearance vary significantly. For example, during rapid target motion, the event count increases sharply. If the neuron's maximum emitted integer value is fixed, the conveyed information becomes limited, potentially resulting in the loss of critical temporal cues. Conversely, in low-motion or static scenarios, excessive spike capacity may result in redundant responses, increasing energy consumption and introducing noise. Therefore, we propose a novel Dynamic Integer Leaky Integrate-and-Fire (DI-LIF) neuron model. In DI-LIF, the maximum emitted integer value is increased under dense event input to enhance responsiveness to informative signals, and reduced under sparse conditions to suppress redundancy and improve efficiency. Specifically, given an input feature tensor $X \in \mathbb{R}^{B\times C\times H\times W\times T}$, we first compute the average feature activation across the spatial and temporal dimensions:

$$\mu(X) = \frac{1}{HWT}\sum_{h=1}^{H}\sum_{w=1}^{W}\sum_{t=1}^{T} X[:,:,h,w,t]. \tag{7}$$

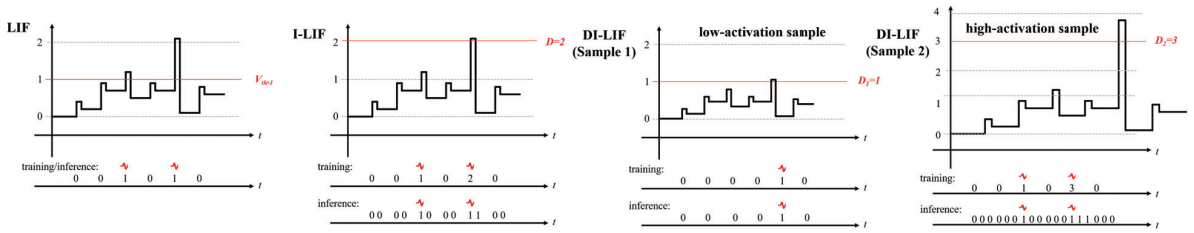


Figure 3. Comparison of LIF, I-LIF, and the proposed DI-LIF neuron. LIF quantizes membrane potentials into binary spikes, causing quantization errors. I-LIF employs integer-valued firing levels but still uses a fixed quantization scale D, neglecting activation diversity. DI-LIF introduces a sample-adaptive dynamic integration mechanism that assigns each sample i a distinct threshold D_i (e.g., D_1=1, D_2=3), enabling refined quantization and balanced inference.

This aggregated representation is then linearly transformed into a scalar adjustment factor α:

$$\alpha = \sigma(W \cdot \mu(X) + b), \tag{8}$$

where $W \in \mathbb{R}^{d\times 1}$ and $b \in \mathbb{R}$ are the parameters of a learnable linear layer; $\sigma(\cdot)$ denotes the sigmoid function. Finally, the batch-specific quantization depth is computed as:

$$D_{\text{batch}} = \lfloor \alpha \cdot D_{\text{init}} + D_{\text{init}} \rceil, \tag{9}$$

where D_{init} is a predefined base quantization emitted integer value. $\lfloor \cdot \rceil$ denotes the floor function that rounds to the nearest integer. This formulation enables the model to dynamically adjust the maximum emitted integer value based on the statistical characteristics of each input batch, thereby improving computational efficiency while preserving task-relevant information. Figure 3 further provides a comparison of the spike firing patterns among LIF, I-LIF, and our DI-LIF neurons in a batch.

Compared with CNN layers, Transformer layers primarily handle interactions between the template and search frames, thereby introducing high-entropy global information. Therefore, we integrate the DI-LIF model into the Transformer, enabling dynamic adjustment of the neuron's maximum firing count to increase firing density, achieving more stable temporal encoding under high-entropy conditions. We design a DI-LIF-based SNN Transformer block as illustrated in Figure 2(c). Formally, this process can be defined as:

$$\mathbf{U}' = \mathbf{U} + \text{SSA}(\mathbf{U}),\ \mathbf{U}'' = \mathbf{U}' + \text{SNNMLP}(\mathbf{U}'), \tag{10}$$

where SSA$(\cdot)$ denotes the Spiking Self-Attention mechanism and SNNMLP$(\cdot)$ represents a spiking multilayer perceptron. The SSA module facilitates cross-region feature interaction between template and search area tokens via spike-driven correlation. Specifically, input tokens are projected into query, key, and value spike matrices through learnable linear transformations:

$$\mathbf{Q}_s = \mathbf{W}_q\mathbf{U}, \quad \mathbf{K}_s = \mathbf{W}_k\mathbf{U}, \quad \mathbf{V}_s = \mathbf{W}_v\mathbf{U}, \tag{11}$$

where $\mathbf{W}_{q/k/v}$ are weight matrices. The attention-driven feature refinement is formulated as:

$$\text{SSA}(\mathbf{Q}_s, \mathbf{K}_s, \mathbf{V}_s) = \left(\mathbf{Q}_s\mathbf{K}_s^\top/\sqrt{d_h}\right)\mathbf{V}_s, \tag{12}$$

where d_h is the scaling factor. We ultimately employ eight DI-LIF based SNN Transformer blocks to further extract spatiotemporal features from the template and search frames, while enabling effective interaction between them.

3.4. I-LIF Based SNN Tracking Head and Loss

The I-LIF based SNN Tracking Head is composed of three parallel branches for predicting object center heatmaps, size vectors, and spatial offsets [28]. Each branch adopts a four-layer convolutional hierarchy, interleaved with I-LIF spiking neurons that enhance spatiotemporal information flow through spike-based dynamics. The center prediction branch outputs a score map $\hat{C}$, indicating the likelihood of the target center at each spatial location. Concurrently, the size and offset branches regress the object dimensions $\hat{S} = [\hat{w}, \hat{h}]$ and the local displacement $\hat{O} = [\hat{o}_x, \hat{o}_y]$, respectively. All branches follow the same processing pipeline:

$$X_i = \sigma(\mathcal{SN}(f_i(X_{i-1}))), \quad i \in \{1, 2, 3, 4\} \tag{13}$$

where $f_i(\cdot)$ denotes convolution, $\mathcal{SN}(\cdot)$ the I-LIF spiking function, and $\sigma(\cdot)$ the bounded sigmoid normalization. Final predictions are computed via:

$$\hat{C} = \text{Conv}_{1\times1}(X_4^c), \quad \hat{S} = \text{Conv}_{1\times1}(X_4^s), \quad \hat{O} = \text{Conv}_{1\times1}(X_4^o) \tag{14}$$

At inference time, the location $\mathbf{p} = (x, y)$ with maximum response in $\hat{C}$ is used to extract $\hat{SN}_\mathbf{p}$ and $\hat{O}_\mathbf{p}$, and the final bounding box $\hat{B} = [x', y', w, h]$ is given by:

$$x' = \frac{x + \hat{o}_x}{s}, \quad y' = \frac{y + \hat{o}_y}{s}, \quad w = \hat{w}, \quad h = \hat{h} \tag{15}$$

where s is the output stride. This formulation enables efficient and biologically inspired target localization under event-driven paradigms. We use focal loss [12] and GIoU loss [16] for training the network. The total loss L is:

$$L = L_{cls} + \lambda_{iou}L_{iou} + \lambda_{L_1}L_{L_1}, \tag{16}$$

where L_{cls} denotes the weighted focal loss for classification, the generalized GIoU loss L_{iou} and L1 loss L_1 are adopted for bounding box regression, $\lambda_{iou} = 2$ and $\lambda_{L_1} = 5$ are the hyperparameters.

4. Experiments

4.1. Experimental Settings

Implementation Details. SpikeTrack is implemented in PyTorch and trained on eight NVIDIA RTX 4090 GPUs with a batch size of 8. We adopt the AdamW optimizer, employing a weight decay of 10^{-4} and an initial learning rate of 4×10^{-4}. The search sequence length is set to 10, and the maximum emitted integer parameter D_{init} is set to 6. We train the model for 60 epochs, each comprising 20K samples. Input frame dimensions for the search and template frames are 256×256 and 128×128, respectively. Our MSST with ten search frames incurs modest training overhead (5.8 GB $\rightarrow$ 18.8 GB; 2.5 h $\rightarrow$ 8 h) but substantial performance gains over a single-frame baseline.

Datasets. To comprehensively evaluate the effectiveness of our SpikeTrack, we conduct detailed comparisons with state-of-the-art (SOTA) trackers on three representative event-based benchmarks: VisEvent [20], FE108 [31], and FELT [21]. VisEvent is a large-scale benchmark containing 500 challenging sequences annotated with 17 attributes for robust event-based tracking evaluation. The FE108 benchmark involves multimodal tracking tasks across various indoor environments, comprising 78 training and 32 test sequences. The FELT provides a challenging dataset for evaluating single-object tracking in long-term scenarios, featuring 742 videos with extensive RGB and event data.

Evaluation Metrics. We employ two standard metrics for quantitative tracking performance evaluation: Success Rate (SR) and Precision Rate (PR). SR measures the proportion of frames where the overlap between predicted and ground-truth bounding boxes exceeds a predefined threshold. PR evaluates the percentage of frames whose center distance between the predicted and actual bounding boxes is within a certain threshold. Additionally, we measure energy consumption for Artificial Neural Networks (ANNs) and SNNs using the following equations:

$$E_{ANN} = O^2 \times C_{in} \times C_{out} \times k^2 \times E_{MAC}, \quad (17)$$

$$E_{SNN} = (T \times D) \times fr \times O^2 \times C_{in} \times C_{out} \times k^2 \times E_{AC}, \quad (18)$$

where O represents the feature map dimension; C_{in} and C_{out} indicate the number of input and output channels, respectively; k is the kernel dimension; fr is the mean spike firing rate; T denotes the timestep count; D represents the upper bound of integer activation values. Consistent with standard practice in energy consumption analyses within the SNN community [13, 15, 26, 29], we assume all computations use 32-bit floating-point arithmetic on 45 nm technology nodes, with $E_{MAC} = 4.6$ pJ and $E_{AC} = 0.9$ pJ [11]. As shown in Eq. (17) and Eq. (18), the reduced energy usage of SNNs is mainly attributed to their sparse spiking activity, resulting in fewer arithmetic operations.

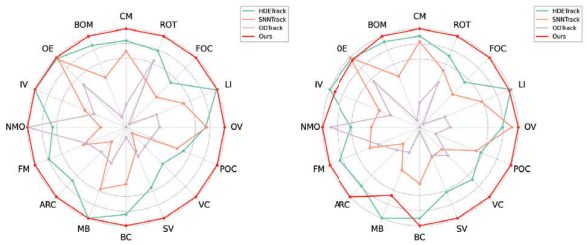


Figure 4. Comparison of SR (left) and PR (right) across various attributes using ours against top trackers on VisEvent.

4.2. Comparisons with State-of-the-arts

The comprehensive tracking performance is illustrated in Table 1, demonstrating that the proposed SpikeTrack provides state-of-the-art tracking performance across all three datasets. Specifically, on the VisEvent [20] dataset, SpikeTrack outperforms the runner-up by 1.6% and 1.8% in SR and PR, respectively. On the FE108 [31] dataset, the proposed approach achieves a 92.7% overall PR and 60.3% SR, and it outperforms the runner-up by 0.5% and 0.5%, respectively. Finally, on the FELT [21] dataset, the proposed SpikeTrack still remains the best performer in SR and PR. This result demonstrates the effectiveness of our method in long-term tracking. Notably, compared to the second-best method HDETrack, our approach uses only 25.8% of its parameters and 6.6% of its energy consumption. Furthermore, although SNNTrack also adopts a spiking neural network architecture, our SpikeTrack exhibits lower parameter count and energy usage, while achieving significantly better tracking performance across all three datasets. These results collectively demonstrate that SpikeTrack effectively balances accuracy, efficiency, and robustness, making it well-suited for real-time event-based tracking applications under resource-constrained and dynamic environments. Addressing real-world adverse conditions is a primary motivation for employing event-based cameras in object tracking tasks. We thus present the tracking performance of the four leading approaches on VisEvent and FELT datasets in sixteen degraded circumstances. As shown in Figure 4, our approach performs best in most degraded conditions, with Figure 5 further qualitatively demonstrating its advantages in two specific scenarios: fast motion and partial occlusion.

4.3. Ablation Study and Analysis

Impact of DI-LIF Neuron. The DI-LIF neuron is a key component of our SpikeTrack, designed to dynamically ad-

Methods	Source	Timestep training/inference	Parameters (M)	Power (mJ)	FE108		FELT		VisEvent	
					SR(%)	PR(%)	SR(%)	PR(%)	SR(%)	PR(%)
ATOM [8]	CVPR19	1×1 / 1×1	-	-	52.8	80.0	22.3	28.4	29.1	46.2
DiMP [1]	ICCV19	1×1 / 1×1	-	-	53.4	88.2	37.8	48.5	32.2	43.4
PrDiMP [9]	CVPR20	1×1 / 1×1	-	-	55.2	86.8	34.9	44.5	32.7	45.7
STARK-S [25]	ICCV21	1×1 / 1×1	28.23	58.88	55.4	83.7	39.6	51.7	34.8	47.0
TransT [4]	CVPR21	1×1/ 1×1	-	-	56.7	89.0	35.2	45.1	32.9	47.1
OSTrack [28]	ECCV22	1×1 / 1×1	92.52	98.90	54.3	86.2	35.9	45.5	33.7	45.3
MixFormer [7]	CVPR22	1×1 / 1×1	37.55	-	57.1	88.6	38.9	50.4	-	-
ARTrack [23]	CVPR23	1×1 / 1×1	202.56	174.80	56.6	87.4	39.5	49.4	32.3	42.8
SeqTrack [5]	CVPR23	1×1 / 1×1	90.60	302.68	52.2	83.8	33.0	42.0	25.4	39.5
HIPTrack [2]	CVPR24	1×1 / 1×1	120.41	307.74	50.8	81.0	38.2	48.9	32.3	42.8
ODTrack [36]	AAAI24	1×1 / 1×1	92.83	335.80	43.2	69.7	29.7	35.9	30.8	41.1
SNNTrack [34]	TIP25	5×1 / 1×1	31.40	8.25	57.2	89.0	-	-	35.9	49.1
HDETrack [22]	CVPR24	1×1 / 1×1	97.82	120.8	59.8	92.2	-	-	37.3	52.5
SpikeTrack	Ours	10×D / 1×D	**25.26**	**7.92**	**60.3**	**92.7**	**41.0**	**52.3**	**38.9**	**54.3**

Table 1. Quantitative comparison of Parameters / Power / SR(%) / PR(%) against SOTA trackers on three event-based tracking benchmarks. The best two results are highlighted in **bold** and underline. D denotes the maximum emitted integer value.

just the maximum emitted integer value based on input features. To evaluate its effectiveness, we conducted two ablation studies: (i) replacing DI-LIF with a conventional I-LIF neuron, and (ii) replacing DI-LIF with an LI-LIF neuron, where the maximum emitted integer value is treated as a learnable parameter. Correspondingly, we obtain two retrained models. As shown in Table 2 #2, replacing DI-LIF with I-LIF neurons leads to a 0.8% and 1.3% drop in SR and PR, respectively, indicating that dynamic threshold adjustment contributes to better tracking performance. When using LI-LIF neurons, SR and PR decrease by 0.5% and 1.5%, respectively. This is because the learnable maximum emitted integer value is fixed at inference, preventing adaptive adjustment to varying inputs.

#	Method	SR(%)	PR(%)
1	**Ours**	**38.9**	**54.3**
2	DI-LIF → I-LIF	38.1	53.0
3	DI-LIF → LI-LIF	38.4	52.8
4	Depth 8 → 4	37.4	52.0
5	Depth 8 → 12	38.0	52.2
6	Center Head → Corner Head	37.0	51.9
7	Center Head → Liner Head	35.3	49.1

Table 2. Ablation studies on the VisEvent dataset.

Impact of Transformer Layers and Tracking Head. To assess the impact of architectural choices on tracking performance, we conduct ablation studies focusing on transformer depth and tracking head design. Varying the number of transformer layers (rows #4 and #5), we observe that reducing the depth from 8 to 4 leads to a clear performance drop, indicating insufficient feature extraction capacity. Increasing the depth to 12 results in a slighter decline, suggesting that excessive depth may introduce redundancy or overfitting. We also evaluate the effect of tracking head designs (rows #6 and #7) by replacing the center-based head with corner-based and line-based alternatives. Both lead to performance degradation, particularly with the line-based head, highlighting the center head's advantage in providing robust and accurate localization for event-based data.

Length	SR(%)	PR(%)	Length	SR(%)	PR(%)
1	34.6	48.1	**10**	**38.9**	**54.3**
2	36.8	50.8	12	37.1	51.8
5	37.7	52.3			

Table 3. Impact of the length of search sequence.

Influence of Search Sequence Length. Using multiple search event frames is a key training strategy tailored to our SNN architecture. The sequence length directly affects tracking performance by controlling the amount of temporal context, which is critical for motion modeling and spike-based feature accumulation in SNNs. To assess this effect, we conducted ablation experiments with varying sequence lengths. As shown in Table 3, short sequences result in suboptimal performance due to limited temporal context. Performance improves with moderate lengths and peaks at 10 search frames, where the model effectively captures spatiotemporal patterns. Further increasing the length to 12 frames leads to a slight decline, indicating diminishing returns and potential redundancy. These results underscore the need to balance temporal integration with computational efficiency in SNN-based tracking.

D_{init}	SR(%)	PR(%)	D_{init}	SR(%)	PR(%)
4	36.6	51.1	8	36.6	50.9
5	37.0	51.2	9	36.4	50.0
6	37.2	**51.5**	10	**37.3**	50.8
7	36.8	50.8	12	36.9	50.6

Table 4. Impact of quantization max_value D_{init} .

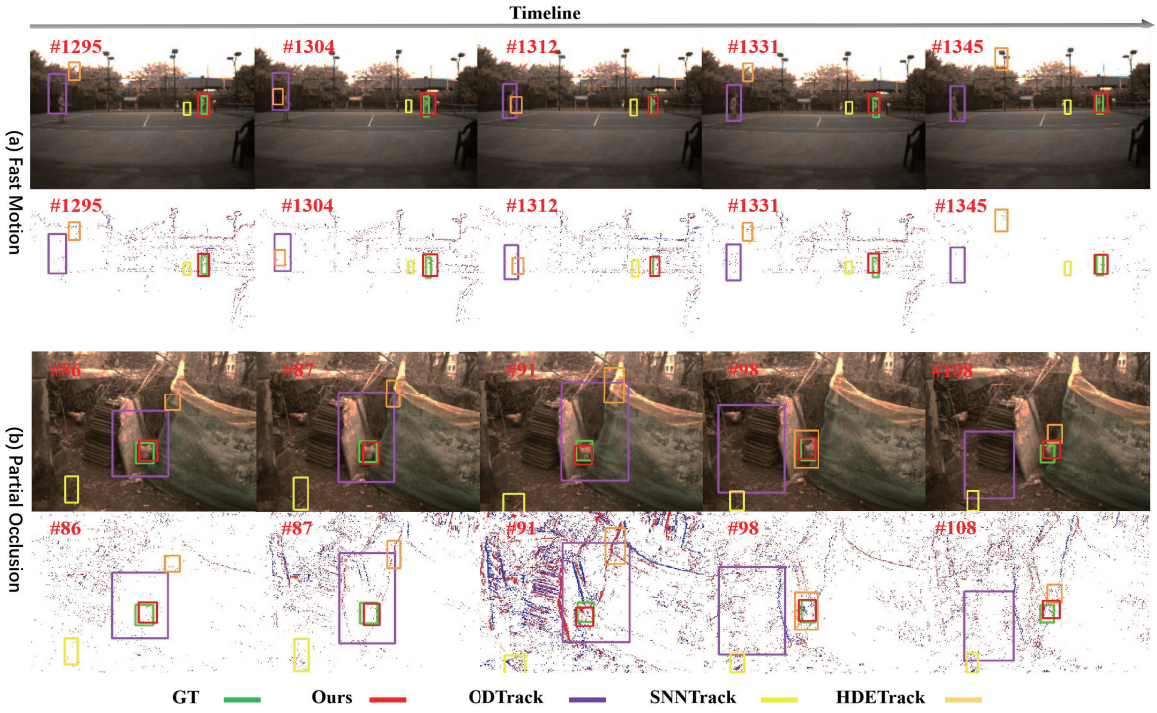


Figure 5. Visual comparison of our SpikeTrack against other SOTA trackers two challenging conditions.

Impact of Initialization of D_{init}. The initialization of D_{init} primarily affects the maximum emitted integer value, thereby modulating the maximum emitted frequency in the SNN. To assess the influence of D_{init} on tracking performance, we assign it a value ranging from 4 to 12. As shown in Table 4, tracking performance remains stable across different values, with SR varying by only 0.9% and PR by 1.5%, indicating low sensitivity to this parameter. This robustness is attributed to our scalar adjustment factor α, which dynamically adjusts the threshold based on input features, enabling not only reduced sensitivity to initialization D_{init} but also adaptive control over spike firing frequency. As a result, our proposed SpikeTrack can maintain consistent tracking performance across different conditions.

Method	Metric	Test Sequence length				
		300	500	700	1000	2000
HDETrack	SR(%)	24.9	24.6	23.4	24.7	24.5
	PR(%)	37.3	36.2	34.1	36.0	34.0
Ours	SR(%)	24.9	24.9	24.8	24.8	26.0
	PR(%)	36.5	36.7	35.5	36.8	35.9

Table 5. Performance comparison of SOTA methods across varying test sequence lengths on the VisEvent Dataset.

Effectiveness for Long-term Tracking. Table 5 presents a comparison of our method with HDETrack on long-term event-based tracking. Although HDETrack performs competitively on short sequences (*i.e.*, 300 frames), our method demonstrates superior performance as the sequence length exceeds 500 frames. Notably, at 2000 frames, our approach outperforms HDETrack by absolute margins of 1.5% in SR and 1.9% in PR. These results highlight the ability of our SNN-based architecture to capture temporal dependencies for robust and accurate long-term tracking.

5. Conclusion

We present a novel pure spike-SNN network for event-based object tracking tasks, SpikeTrack, that is capable of adaptively extracting spatial and temporal information encoded in the events. Our proposed SpikeTrack adopts a multi-frame search training strategy, which aligns well with the temporal modeling capabilities of SNNs and the inherent reliance of object tracking on temporal continuity and motion consistency. Moreover, our proposed DI-LIF neuron adaptively performs integer-based training and spike-driven inference based on input features, effectively reducing quantization error and enhancing generalization across diverse scenarios. Experiments on FE108, FELT, and VisEvent show our method achieves SOTA accuracy with reduced power, highlighting spiking computation's benefits. We hope SpikeTrack will inspire further research into efficient, high-performance SNNs for real-world vision tasks.

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